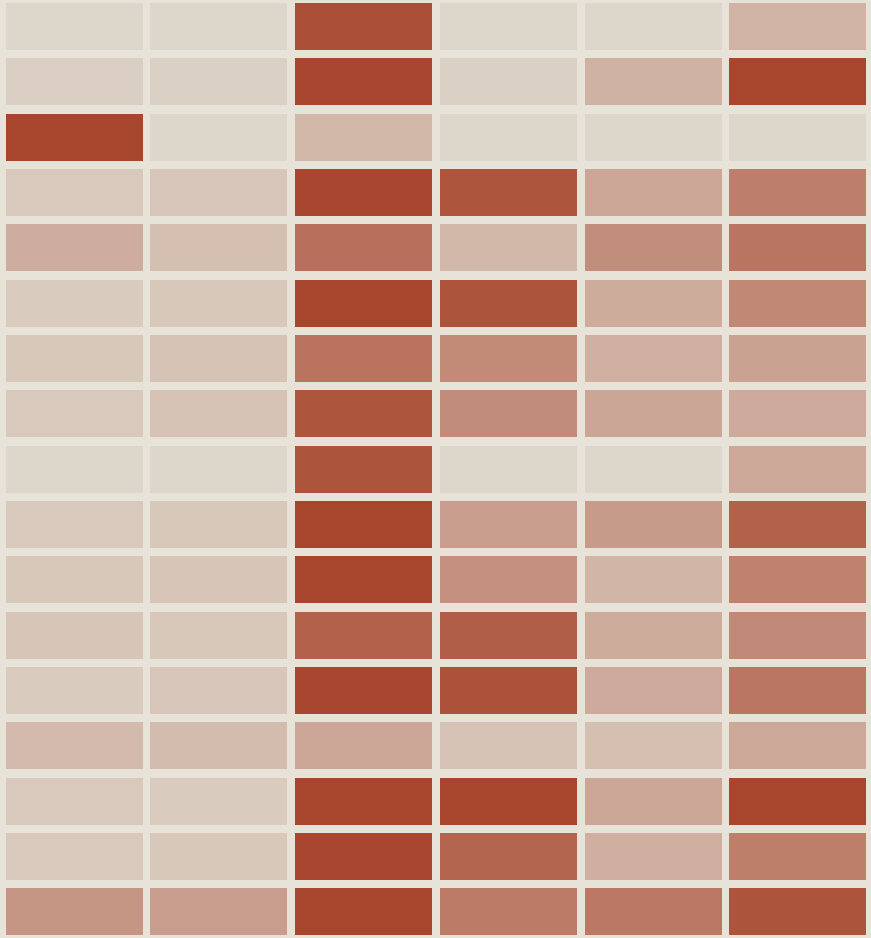
Anomaly detection for data quality, measured inside the ETL pipeline

Living progress deck. Rule-based, classical and first deep detectors run through one pipeline and are compared per anomaly type, on a synthetic generator and on 42 real UCI tables with injected anomalies, and on 100 labelled problems of the pool.

Author	Theo Bigand, 2026-11-11
Repository	theobigand/etl-anomaly-benchmark
Evidence	pool of 152 open sets, synthetic generator, 17 detectors



Mean recall, 17 detectors by 6 anomaly types, synthetic customer table, 5 seeds. Rows: Z-score to LUNAR as in the paper. Each family has a narrow band of dark cells.

Does machine learning in an ETL pipeline beat plain rules, and for which anomalies?

The industry default is rule-based: auditable, cheap, and blind to what the author did not anticipate. Recent work adds ML detectors inside the flow, but the evidence is thin, one algorithm on one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

What generic anomaly benchmarks leave out

EVALUATED IN A VACUUM

Algorithms scored outside any pipeline, no integration cost measured.

NO DECLARATIVE BASELINE

The rules that dominate production are never in the comparison.

ONE AGGREGATE SCORE PER DATASET

Hiding which family is good at which kind of defect.

A SINGLE FLAG THRESHOLD

Reported as one operating point, not the precision-recall trade-off.

Benchmarks rank detectors. Few report their decisions

References were checked against a primary source or the DOI registry, with the limits listed in references.md. In the texts I read, ADBench and AnoShift report AUC only, and none of the works below breaks results down by data-quality fault. The scan is not systematic, so this is a gap I did not find filled, not a proof that nothing exists.

Work	Time order	At a threshold
ADBench, 30 algorithms, 57 sets	not found	not found (AUC)
Agyemang 2024, one simulated set	none	precision, recall, F1
AnoShift, Kyoto 2006-2015	IID, NEAR, FAR	not found (AUC)
FedAD-Bench, federated deep	not found	F1-optimal threshold
Sun et al. 2024, online thresholds	shift in a stream	error-rate guarantees

"Not found" means absent from the text I read, not proven absent. This thesis: six fault types, a controlled generator and real tables, decisions and pipeline cost.

4 RESEARCH QUESTIONS

Four questions, one study each, hypotheses written first

Every study has a protocol and hypotheses committed before its runs. A hypothesis that fails is reported as failed: three of the six on the whole pool do.

Question	Studies	Answer
RQ1 rules, classical, neural in one pipeline	examples, 100 labelled problems	historical ranking; train-only replay pending
RQ2 which anomaly type each family finds	synthetic matrix, schemas, 42 UCI tables	narrow competence, values do not transfer
RQ3 operating point and time	contamination, label-free threshold, chronological split	threshold fixed on the past fails when prevalence moves
RQ4 cost in the pipeline	cost study, ETL run on 152 sets	neural 10 to 100 times column-wise

`The Results section follows these four questions; the Discussion answers them in one paragraph.`

A controlled dataset that labels six anomaly types separately

12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall **and** precision are reported per type. The magnitudes are one **severity** multiplier, so the choice is swept rather than fixed.

6

anomaly types, labelled and measured separately

MISSING VALUE	one or two feature cells set to null
EXACT DUPLICATE	a clean row copied, same business attributes
UNIVARIATE OUTLIER	one numeric field pushed far out of range
MULTIVARIATE OUTLIER	every field in range, the combination not (an 18-year-old with a top-percentile balance)
TEMPORAL DRIFT	recent-signup rows whose income and balance distribution has shifted up
INVALID FORMAT	malformed e-mail, out-of-domain country code, impossible age

Because the category of every injected row is known, the benchmark reports a recall **and** a precision per type: the part that generic anomaly benchmarks, which publish one aggregate score, do not provide.

One interface, one measured path

The demo and every benchmark call the same `detect()` stage, so latency is the real in-pipeline fit + predict + score time. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and AP computable; `detect(fit_df=...)` fits on the train slice and scores the test set. Exact deduplication belongs to the ordinary transform; typed injected duplicates instead remain in their separate evaluation with `transform_dedup`.

extract → transform → **detect** → load

```python
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:   # bool per row
        ...
    def score(self, df) -> Series:     # graded score
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector):
    def fit(self, df):
        x = impute(df[self.numeric_cols])
        self.model = self.estimator.fit(scale(x))
        return self

# registry: one line, picked up everywhere
"isolation_forest": lambda: IsolationForestDetector(),
```

6A METRIC INTERPRETATION

Average Precision measures the ranking

AP uses continuous anomaly scores. Precision, recall, F1 and the flagged share use a selected decision threshold.

$$AP = \sum \Delta R \cdot P$$

precision weighted by each recall increment

NON-INTERPOLATED

Distinct cutoffs run from high to low scores; tied scores enter together. AP differs from trapezoidal precision–recall area.

PREVALENCE MATTERS

Prevalence is a no-skill reference precision. Constant scores give AP equal to prevalence; a random strict ranking has an upward finite-sample correction.

SEPARATE DECISIONS FROM SCORES

Changing only the flag threshold leaves AP and ROC-AUC unchanged. ROC-AUC is complementary; neither ranking metric establishes a useful operating point.

HISTORICAL NAMING

Stored `pr_auc` fields and remaining AP labels in archived figures and the paper denote this same AP quantity.

7 THREE FAMILIES, ELEVEN ROW-LEVEL DETECTORS

Rules and classical models, one contract

Classical detectors use PyOD or the recorded protocol adaptations. The flag threshold uses training prevalence. They receive numeric columns only; Great Expectations separately checks nulls on all columns, so this is an explicit information asymmetry. ECOD, COPOD and HBOS aggregate marginal evidence without explicitly fitting dependence between columns. The time-aware `drift` detector is opt-in, synthetic only.

Detector	Family	Principle	Main limit
Z-score	rule	flag `|z| > 3` on any column	mean and std inflated by outliers
IQR	rule	Tukey fences, `k = 1.5`	fires often in high dimension
Great Expectations	rule	null checks + fitted 0.1% / 99.9% bounds	generic statistical suite, no domain rules
Isolation Forest	joint	isolation depth under random splits	axis-aligned splits, weak on nulls
LOF	joint	local density vs 20 neighbours	cost grows with the scored set
One-Class SVM	joint	RBF boundary around the bulk	quadratic training
ECOD	column-wise	empirical tail probabilities, summed	no explicit cross-feature dependence
COPOD	column-wise	max of skew term and mean log tails	same
HBOS	column-wise	histogram per column, inverse bin height	independent columns, coarse bins
KNN	joint	distance to the 5th neighbour	quadratic, skipped above 50k rows
PCA	joint	reconstruction error, 95% of variance	misses a shift inside the kept space
drift	time-aware	recent-window mean shift	per row, so diluted drift is invisible

7A FITTED RULES

Column bounds do not fix row alert rates

All statistics come from training data. The evaluated Great Expectations suite is a generic statistical baseline, not a domain-informed rule set.

Z-SCORE

Maximum |x − training mean| / sample standard deviation. Flag above 3; constant columns contribute zero.

IQR

Training fences Q1 − 1.5 IQR and Q3 + 1.5 IQR. Flag strictly outside; zero-IQR columns contribute zero.

GAUSSIAN REFERENCE ONLY

Per-column tails: 0.27% beyond 3σ, 0.70% beyond the IQR fences (about 2.7σ). Fitted bounds and multiple columns change row-level rates.

GREAT EXPECTATIONS

Not-null on all columns; inclusive numeric training quantiles 0.1% / 99.9%. Omit undefined or equal bounds. Score: fraction of active checks violated.

8 SIX NEURAL MODELS, OPT-IN

Neural detectors behind the same contract

PyOD 3.6.7 with recorded adaptations, CPU, seeded, fitted on at most 20,000 rows. Frozen representation scaling and latent-mean scoring change the evaluated algorithms; they do not establish equivalence to upstream defaults.

Detector	Principle	Fixed or noted
Autoencoder	rebuild standardized rows through a bottleneck	MLP from scikit-learn
VAE	Gaussian code, rebuild, reconstruction error	scored with the code mean, PyOD draws one
Deep SVDD	map normal rows near a centre, distance	weakest on most types
DIF	isolation forests on random neural features	training representation scaler reused
AE1SVM	autoencoder + random-Fourier one-class SVM	singleton padding changes training weight
LUNAR	network learns the score from neighbour distances	best on drift, many false alarms

The LSTM autoencoder reads windows of ordered rows and stays outside the registry.

8A EVALUATED VARIANTS

Stable row scores require explicit adaptations

A fitted ETL detector must score a row without consulting other evaluation rows. Contract tests check the 17 row-level detectors; the time-window monitor has a different contract.

ECOD AND COPOD

PyOD 3.6.7 recomputes tails and skew with the evaluated batch. Here ranks and skew signs are frozen from training. Even singleton scores can differ when the skew sign changes.

COPULA MOTIVATION VERSUS IMPLEMENTED SCORE

COPOD adds marginal contributions: max(skew-selected log tail, mean of left/right log tails). This is not a fitted general joint copula. ECOD takes the max of both log tails and the skew term.

NEURAL SCORING

VAE decodes the latent mean; DIF reuses a training representation scaler. These are algorithm changes. AE1SVM training padding also reweights its last row.

EVIDENCE BOUNDARY

Bounded tests support batch independence, not universal equivalence. Separate ablations remain open; historical results retain their actual adapted configurations.

8B EXPLORATORY CORRECTION

Historical pool preparation needs revalidation

Column filtering, category vocabulary and conflicting-label removal occur before the historical split. Training-only medians and scaling do not isolate those earlier decisions.

CONSTRUCTED NEGATIVE CONTROLS

Changing only evaluation categories changes encoded width from 5 to 3. An evaluation duplicate's label can also remove a reference row. AP bias is not yet quantified.

VERSIONED PATH

pool-train-only-v1 consumes source-bound frozen positions. Fitted schema, imputation and scaling use training rows only. Cross-partition duplicates are rejected; evaluation rows are retained.

BOUNDED CHECKS

Separate constructed checks enforce full line/branch coverage for the two new modules and detect three leakage defects. Targets are excluded from preparation; only training prevalence calibrates flags.

STILL OUTSTANDING

Historical memberships were recomputed (not originally archived) and replay-validated: the Z-score reproduces all 100 archived per-seed results. Replaying the studies on the train-only path remains outstanding; old protocols and results are preserved and no corrected pool-performance claim is available.

Three example datasets bracket the method

Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

0.01 → 0.20

Isolation Forest fraud precision, from the contamination change alone

Dataset	Rows	Feat.	Anom.	Role
Credit Card Fraud	283,726	30	0.17%	rare, but a mixed-defect label
Cardio	1,831	21	9.6%	rare and outlying, the fitting regime
Water Quality	3,276	9	61%	target class common, wrong regime
Synthetic per-type	12,360	9	17.5%	controlled, 6 labelled types

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

10 HOW EACH TYPE IS INJECTED

Magnitudes are a swept parameter, not a fixed choice

severity = 1.0 is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. The severity sweep (slide 15) turns "we picked the magnitudes" into a curve and shows the per-type ranking is stable. Five types replace clean rows; duplicates are appended.

MISSING VALUE	1–2 of {income, balance, avg amount, tenure, n txn, email, country} → null
DUPLICATE	exact copy of a sampled clean row, id included
UNIVARIATE	one numeric column × U(15, 45) × severity
MULTIVARIATE	marginals at the (1 − 0.03 / severity) percentile of the clean data
TEMPORAL DRIFT	signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount
INVALID FORMAT	email → malformed; age → impossible; country → out of domain

At `severity = 1` these are the reference magnitudes. Sweeping the multiplier (slide 15) shows every distributional detector's recall on univariate, multivariate and drift rises smoothly, and the ranking does not change.

Aggregate scores do not establish equivalence. Per-type results show different competencies.

F1 spans 0.22 to 0.59 over the seventeen detectors on the synthetic set · the per-type competences barely overlap

The operating point decides the outcome

The statistical rules recover most frauds at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07). Great Expectations leads AP.

0.20
Isolation Forest precision (was 0.01 at contamination 0.10)

0.182
Great Expectations AP, the best on this dataset

Detector	Prec	Rec	F1	ROC	PR	Flag%	Time
Z-score	0.011	0.870	0.021	0.926	0.078	13.4	0.03s
IQR	0.003	**0.966**	0.007	0.911	0.018	48.6	0.12s
Great Expectations	0.037	0.676	0.070	0.830	**0.182**	3.1	1.37s
Isolation Forest	**0.202**	0.206	**0.203**	**0.934**	0.120	0.2	1.78s

Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

When anomalies are rare and outlying, the learned detectors work

Isolation Forest reaches F1 0.62 and AP 0.70, with ECOD, COPOD and DIF close behind (AP 0.69, 0.69, 0.67). Z-score also does well: the 21 features are standardized and anomalies stand out on several at once. This is the counterpoint to a 0.17% mixed label and a 61% majority.

0.62
Isolation Forest F1 on Cardio

0.70
Isolation Forest AP on Cardio

Detector	Prec	Rec	F1	ROC	PR	Flag%
Z-score	0.49	**0.82**	0.61	0.93	0.61	16.6
IQR	0.21	0.90	0.34	0.78	0.30	43.4
Great Expectations	0.51	0.21	0.29	0.59	0.19	4.6
Isolation Forest	**0.64**	0.62	**0.62**	**0.95**	**0.70**	9.9
LOF	0.25	0.27	0.26	0.62	0.24	10.7
One-Class SVM	0.56	0.61	0.58	0.94	0.63	10.7

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

Aggregate scores hide different trade-offs

Z-score and HBOS reach a perfect precision by firing only on extreme rows (recall 0.20). DIF has the best F1 and AP, the VAE matches Isolation Forest, LUNAR has the best recall with a low precision.

0.22–0.59

F1 range across the seventeen detectors

Detector	Prec	Rec	F1	ROC	PR
RULES AND CLASSICAL					
Z-score	**1.000**	0.196	0.328	0.742	0.544
IQR	0.584	0.396	0.472	0.675	0.452
Isolation Forest	0.495	0.499	0.497	0.735	0.485
One-Class SVM	0.478	0.485	0.481	0.722	0.520
ECOD	0.341	0.341	0.341	0.675	0.283
HBOS	**1.000**	0.204	0.339	0.602	0.343
KNN	0.466	0.469	0.467	0.745	0.520
NEURAL					
VAE	0.502	0.507	0.504	0.742	0.548
DIF	0.589	0.582	**0.585**	**0.772**	**0.648**
LUNAR	0.345	**0.668**	0.410	0.728	0.495

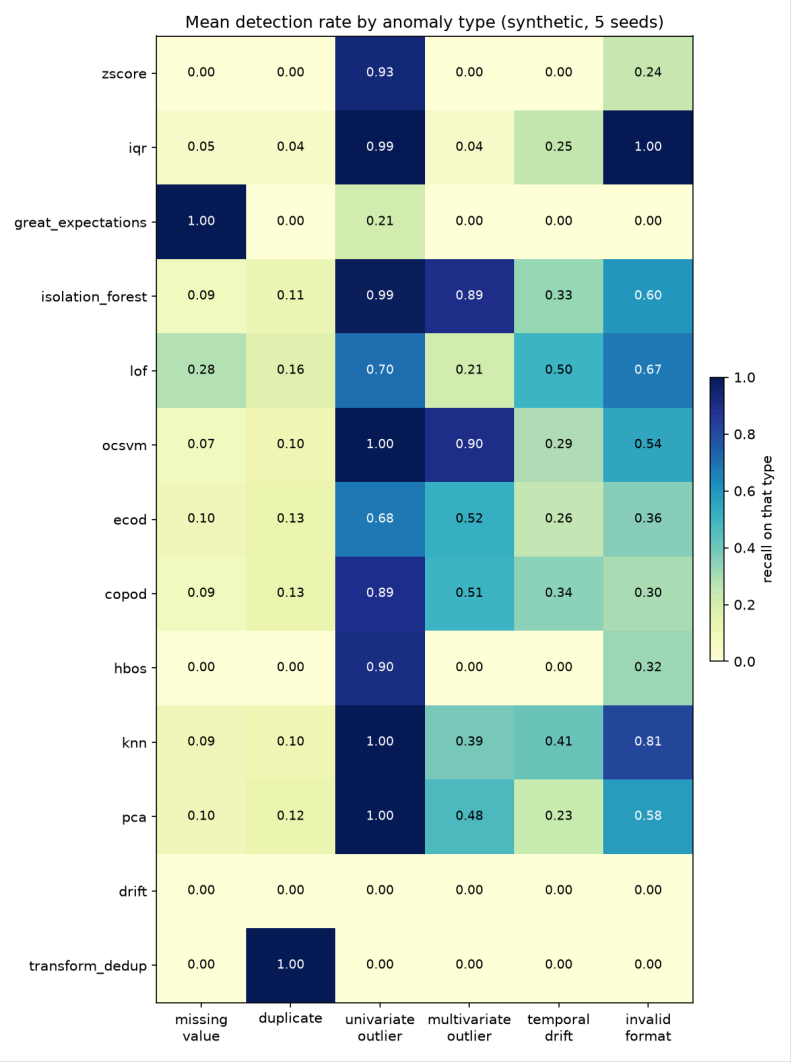
Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous. All 17 detectors are in the paper.

Each method has a narrow competence

Z-score owns univariate outliers. Great Expectations owns missing values (recall 1.00, precision 0.90). Deduplication owns exact duplicates. Isolation Forest and One-Class SVM are the only methods that catch multivariate outliers. Broad coverage needs several families, not the single best one.

0.91 vs ≤0.02

Isolation Forest recall on multivariate outliers, against every rule



paper/figures/heatmap_synthetic.png

transform_dedup and drift shown for reference

16 RECALL AND PRECISION, PER TYPE

Detection rate alone can be gamed

Isolation Forest catches 91% of multivariate outliers at only 0.24 precision: a wide net that also pulls in clean rows. Z-score fires only when it is certain, so its precision is 1.00 on the two columns it covers.

0.91 / 0.24

recall / precision on multivariate outliers, Isolation Forest

Method	missing		univ. out.		multiv. out.		drift	
	rec	prec	rec	prec	rec	prec	rec	prec
Z-score	0.00	–	**0.92**	**1.00**	0.00	–	0.00	–
IQR	0.05	0.03	**0.99**	0.39	0.02	0.01	0.27	0.15
Great Expectations	**1.00**	**0.90**	0.23	0.70	0.00	–	0.00	–
Isolation Forest	0.09	0.03	**0.99**	0.26	**0.91**	0.24	0.33	0.10
LOF	0.26	0.06	0.65	0.15	0.16	0.04	**0.49**	0.11
One-Class SVM	0.08	0.03	**0.99**	0.25	0.84	0.22	0.34	0.10
transform dedup	0.00	–	0.00	–	0.00	–	0.00	–

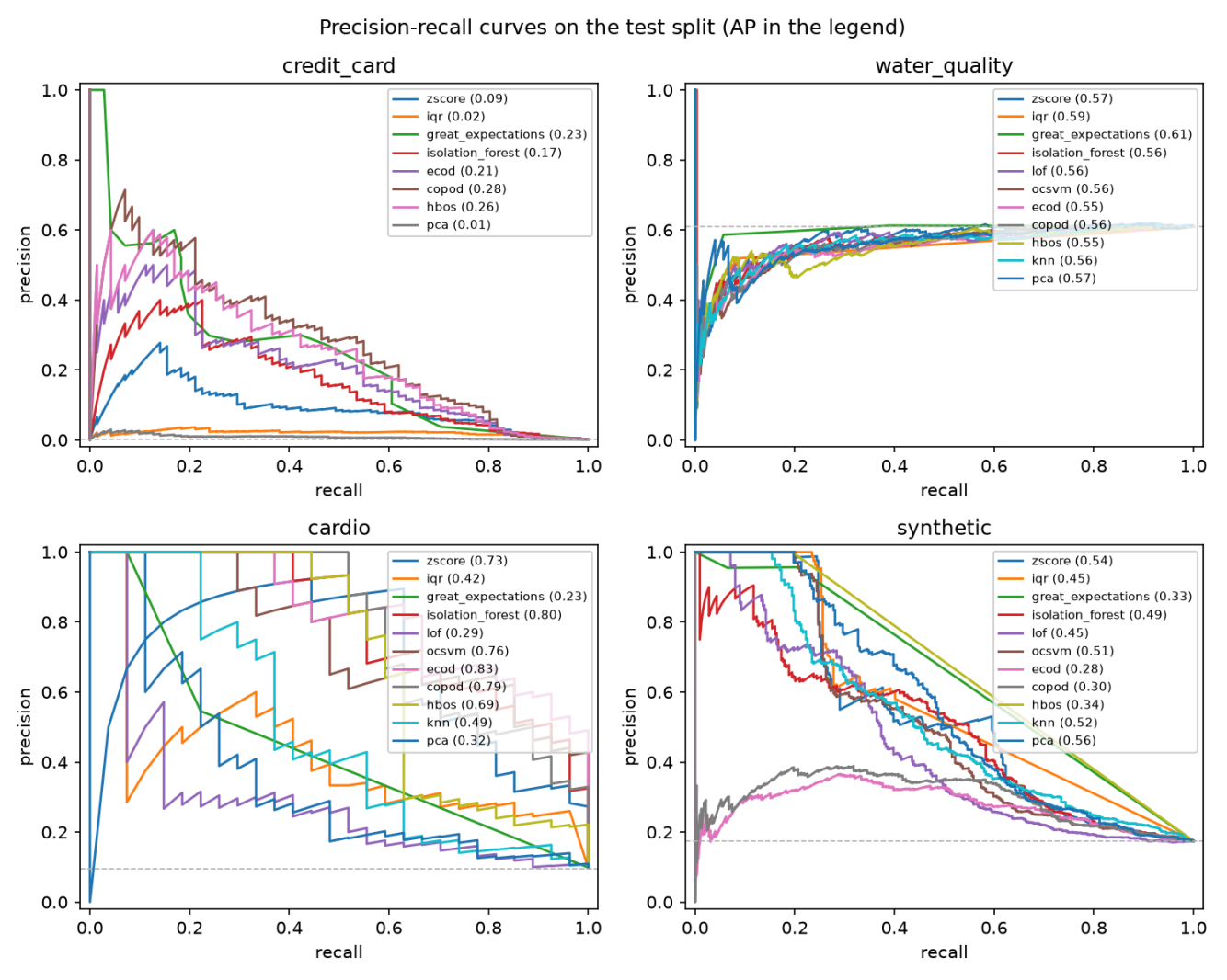
Recall and precision per anomaly type, clean rows as the only negatives. 5 seeds, mean. Duplicate and invalid-format columns omitted for space.

One threshold hides the whole trade-off

On Cardio the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On Water Quality every curve hugs the 0.61 base-rate line. On Credit Card Fraud two methods dominate but neither clears precision 0.4 except at negligible recall.

0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 → 0.92

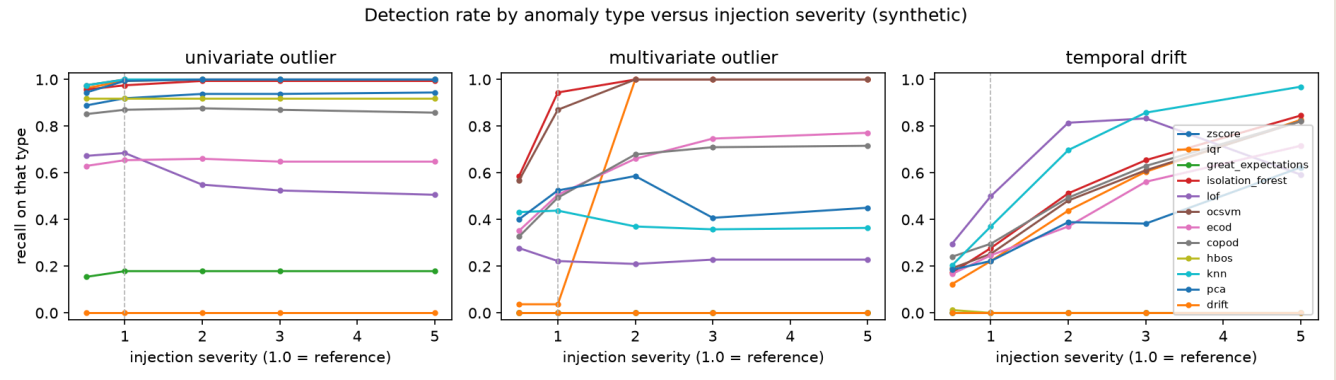


The per-type ranking is stable as the magnitudes scale

Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is simply below every method's floor, not a property of the methods. The same families lead the same columns at every severity.

≤0.3 → 0.85

temporal-drift recall from severity 1 to 3



A pool of 152 open datasets, pinned and checked

Issue #66. The three datasets of M1 bracket the method but are too few to say anything general. The pool follows ADBench in scale and adds a licence rule: a dataset enters only when its terms are read and open. Raw data stays out of the repository; checksums, licences and reports are archived.

152
pinned datasets

116
clean carriers for injection

100
labelled anomaly problems

ADBench tabular sets kept (35 CC BY 4.0, 1 ODbL)	36 OF 47
UCI sets passing metadata and licence filters	133
UCI sets kept after curation, reason recorded	116
anomaly label rule fixed before any detector	RAREST CLASS
one normalisation policy for all sets	SINGLE
download check of every set, manual and monthly CI	PINNED
excluded for unclear terms (celeba, mammography, vowels...)	11

Historical pool rankings need train-only revalidation

Historical run: three rules, eight classical and six neural detectors, five seeds, on 36 ADBench and 64 UCI problems. The numerical table is retained as recorded. Schema and row filtering precede the split, so these descriptive rankings still need train-only replay; the 100 problems come from 80 sources and a source-clustered re-analysis changes no sign (only Great Expectations is clearly below Isolation Forest); label-rule sensitivity and unrecorded PyOD convergence diagnostics remain open.

17
detectors; no neural exception recorded

13
problems won by DIF, the most; each of the 17 wins at least 2

Detector	Problems	AP median [95%]	ROC-AUC	Rank
ECOD	100	**0.240 [0.191, 0.305]**	0.656	7.0
One-Class SVM	84	0.240 [0.177, 0.307]	0.670	8.0
HBOS	100	0.237 [0.176, 0.311]	0.663	8.0
DIF	100	0.226 [0.173, 0.272]	0.639	8.0
VAE	100	0.216 [0.163, 0.279]	0.676	7.0
COPOD	100	0.216 [0.162, 0.314]	**0.691**	7.0
Isolation Forest	100	0.215 [0.158, 0.293]	0.677	7.0
AE1SVM	100	0.210 [0.172, 0.253]	0.635	8.0
Autoencoder	100	0.205 [0.150, 0.240]	0.631	8.0
Z-score	100	0.203 [0.164, 0.279]	0.626	9.0
KNN	92	0.203 [0.172, 0.284]	0.658	8.0
LUNAR	100	0.183 [0.153, 0.225]	0.615	10.0
Deep SVDD	100	0.181 [0.147, 0.226]	0.619	10.0
PCA	100	0.178 [0.152, 0.219]	0.581	10.0
LOF	92	0.165 [0.153, 0.237]	0.588	10.0
IQR	100	0.144 [0.117, 0.191]	0.556	12.0
Great Expectations	100	0.115 [0.105, 0.123]	0.508	16.0

Median over problems, sorted by AP. LOF and KNN are skipped above 50,000 training rows and One-Class SVM above 20,000. Neural models fitted on at most 20,000 rows.

labels from a rare class, not from data-quality defects

Two regimes the three examples do not show

Among the 100 problems, the largest margin of the rules over the learned models is on Annthyroid (IQR 0.61, best learned 0.31). The largest margin of a neural model is on Mushroom (LUNAR 0.96, best classical 0.63; label derived by a rule). Chosen after the results, as illustrations, not as a test.

0.61
IQR rule on Annthyroid (best learned model 0.31)

0.96
LUNAR on Mushroom, where the IQR rule is at chance (0.10)

AP	Credit Card	Cardio	Annthyroid	Mushroom
Z-score	0.10	0.61	0.18	0.44
IQR	0.02	0.30	**0.61**	0.10
Isolation Forest	0.13	**0.70**	0.29	0.63
ECOD	0.21	0.69	0.28	0.41
HBOS	0.26	0.58	0.24	0.50
Autoencoder	0.11	0.35	0.14	0.68
DIF	0.27	0.67	0.23	0.45
LUNAR	0.15	0.33	0.20	**0.96**

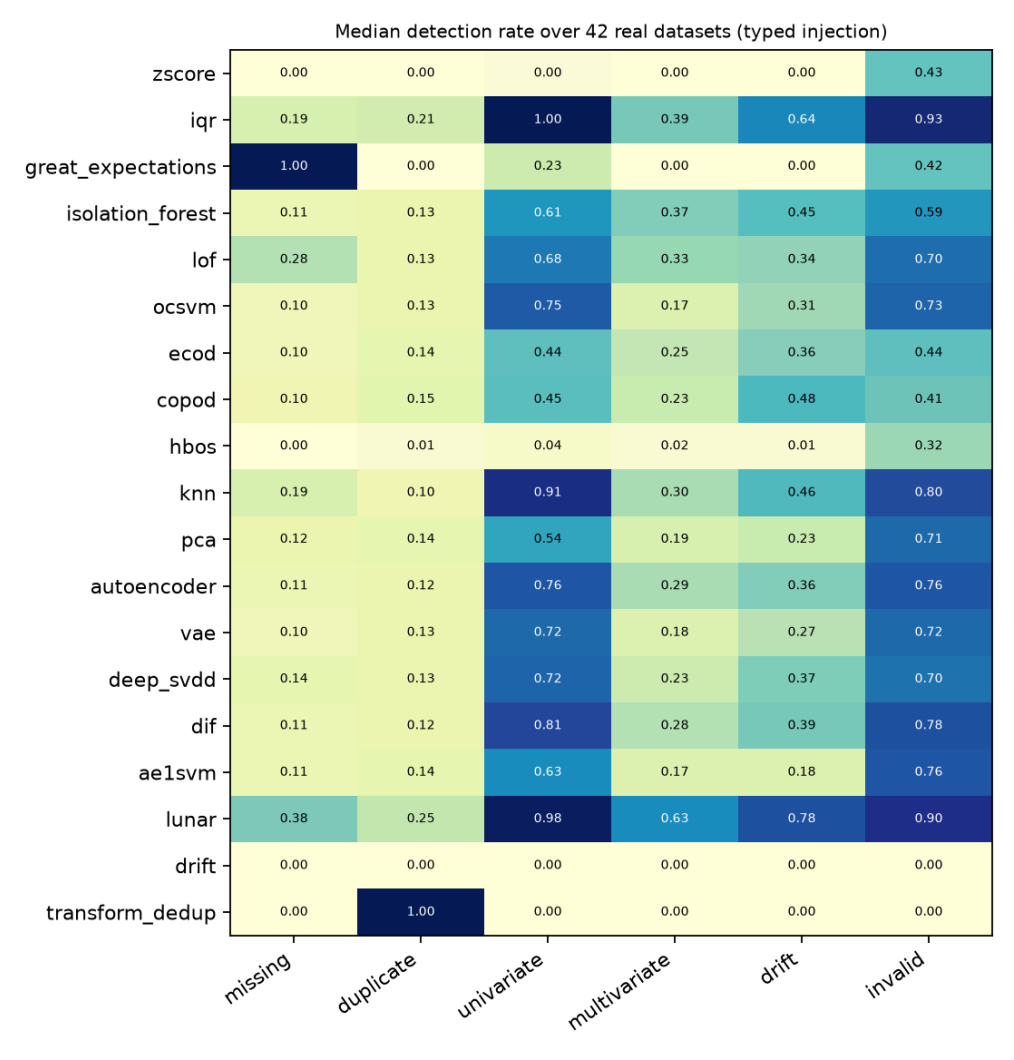
Mean over 5 seeds, as run in the pool (same normalisation for all). Best of each column in green; all 17 detectors are in the paper.

The synthetic pattern does not simply carry over

Issue #65. The six anomaly types are injected into 42 real UCI tables (3% of rows per type, same split and settings as the synthetic run). Protocol and five hypotheses were written before any run. Two rows hold everywhere: Great Expectations finds all missing values, deduplication all duplicates. Three hypotheses are not supported. The eleven detectors added later run on the same tables: LUNAR is best on multivariate outliers (0.63) and drift (0.78), but flags a median 24% of clean rows.

0.37
Isolation Forest, multivariate outliers (0.89 synthetic)

24%
median clean rows flagged by LUNAR (IQR 22%)



median over 42 UCI tables (58 with ADBench, same picture), injected not natural

Two failures that are explained, not hidden

Z-score detects almost nothing on univariate outliers (median 0.00 against 0.93 synthetic). A check made after seeing it, so exploratory, shows why: invalid rows in the training split inflate the standard deviation. Without them the Z-score finds every univariate outlier on all 42 tables. The IQR fences reach the best detection rate on several types, but only by flagging a median 22% of clean rows.

0.83
mean rank correlation between two tables

-0.96
IF detection vs number of columns

MASKING

Non-robust statistics on a contaminated training set, not a weakness of the Z-score as such.

DETECTION RATE CAN BE GAMED

The false positive rate on clean rows has to be read next to it. It reaches 99% on some wide tables.

STABLE RANKING

Rank correlation of the method-by-type matrices is 0.83, lowest for drift (0.70). Partly mechanical: many cells sit near 0 or 1.

DIMENSION HURTS

Isolation Forest on univariate outliers falls as the number of continuous columns grows.

Time changes the answer: a threshold from training does not travel

Credit Card is the only dataset with a clock. Detectors are fitted on the first 70% of time and scored on the last 15% (52 frauds). Average precision drops for all four detectors, but the interval excludes the stratified mean only for Great Expectations and Isolation Forest. Prevalence also changes between partitions, so the pilot does not separate it from the effect of time.

0.936
Isolation Forest ROC-AUC, chronological

0.08%
future rows it flags: none of the 52 frauds

Detector	PR strat.	PR chrono	ROC chrono	Flag%
Z-score	0.078	0.048	0.935	12.8
IQR	0.018	0.013	0.917	47.1
Great Expectations	0.183	0.045	0.758	3.5
Isolation Forest	0.110	0.035	0.936	0.08

Average precision on the test split, 95% block-bootstrap interval for the chronological arm in the paper. The window monitor detects mean shifts of 1 and 2 standard deviations, not 0.5.

Part of the gap is only the prevalence

Issue #63. Credit Card and Bank Marketing (UCI, ordered by date 2008 to 2010), four detectors, three label-free thresholds. On Bank Marketing the prevalence goes from 5.8% in training to 39.3% in test: matching the stratified control to it removes most of the gap. A rolling threshold that keeps the training prevalence cannot follow it. One hypothesis of four holds.

0.26 → 0.06

median AP gap on Bank Marketing after prevalence matching

F1 at the threshold	P0 training	P2 rolling
Bank, Z-score	0.51	0.02
Bank, Isolation Forest	0.41	0.11
Credit Card, Z-score	0.00	0.09
Credit Card, Great Expectations	0.00	0.09

Test period, threshold at the training prevalence. Rank correlation of the detector order, stratified against chronological: 0.80 on Bank Marketing, 0.40 on Credit Card. Kyoto-2006+ left out: rights and download not verified.

25A M2 · EXPLORATORY DEPLOYMENT THRESHOLDS

A nominal alert budget is not a promise

Issue #94, exploratory. Budgets of 0.5 to 10% set from training scores only, on 100 labelled problems. Held-out alert volume was 1.05 to 1.36 times the nominal 1% budget for classical detectors and 1.13 to 2.87 times for neural ones. One machine; not in the main tables.

+4.1 budgets

added alert volume at a 2% budget (about 5 times nominal) for a 1 σ score shift; ROC-AUC unchanged

Change at a 2% budget	Δ alert ÷ budget [95%]	Δ ROC-AUC
Evaluated scores shifted +1 σ	+4.09 [3.50, 4.77]	0.000
Evaluated scores scaled ×2	+4.86 [4.57, 5.19]	0.000
Noise of 1 σ on the scores	+1.55 [1.32, 1.81]	−0.038
Every feature drifted ½ σ	+20.90 [17.39, 24.55]	−0.051
Monotone change of fit and test scores	+0.001 [0.000, 0.003]	0.000

Five classical detectors, 98 problems from 78 sources, sources resampled whole, Holm-adjusted. Prevalence reweighted under a label-shift assumption: 0.83 to 1.83 times the budget at 0.5% prevalence, 1.15 to 2.73 times at 20%. The 30-expected-alert rule for monitoring was chosen after the false alarms were seen.

The deeper studies, run on all 152 datasets

Each study ran on every dataset where it is defined; the others are listed with a reason. Six hypotheses were written first. Three hold and three do not, and the failures are reported as such.

58 / 152
tables eligible for typed injection and severity

290 / 304
end-to-end pipeline runs that completed

Hypothesis	Result	Value
P1 label-free threshold lowers F1	**holds**	14 of 17 detectors
P2 curve ranking = AP ranking	fails	Spearman 0.90 (< 0.99)
P3 recall rises with severity	fails	13 of 17; column-wise rules stay at 0
P4 detection share of the pipeline	**holds**	62% (IF), 4% (Z-score), 14 datasets
P5 chronological AP lower	fails	31 of 64 pairs; Bank Marketing 0/16
P6 files match the manifest	**holds**	152 of 152

Chronology needs a real time variable: 5 labelled datasets qualify (ADBench has none).

An LSTM autoencoder is comparable, not clearly better

Issue #52. The window is the ten previous rows of the global stream, since Credit Card has no cardholder. The protocol and a CPU budget of 20 epochs were fixed beforehand. On a synthetic stream with broken continuity the LSTM wins by a wide margin. On the real frauds it only matches Isolation Forest and Great Expectations, and none of five trainings converged within the budget.

0.875
LSTM AP on the synthetic stream (row AE 0.006)

0.037
LSTM AP on Credit Card (row AE 0.024)

paired bootstrap, main seed	+0.021 [+0.005, +0.044]
other four seeds	INTERVAL INCLUDES 0
versus Isolation Forest (0.035), GX (0.045)	COMPARABLE
convergence within 20 epochs	NOT REACHED
recall of frauds in the later period	≤ 0.02
deep models in the per-type matrix	#64 DONE

The profile is stable, two cells are not

Issue #67. Transactions and sensor readings get the same six injected types and the same protocol as the customer table. Great Expectations and the deduplication rule carry over exactly, and the rank correlation between schemas is 0.93. The multivariate column and drift depend on how the columns are scaled and correlated. Two of four hypotheses hold.

0.93
mean rank correlation between schemas

0.96
best drift recall on sensors (0.38 on transactions)

Multivariate outliers	customers	transactions	sensors
Isolation Forest	0.89	0.49	0.32
One-Class SVM	0.90	0.18	0.33
IQR	0.04	0.57	0.02
Z-score	0.00	0.00	0.00

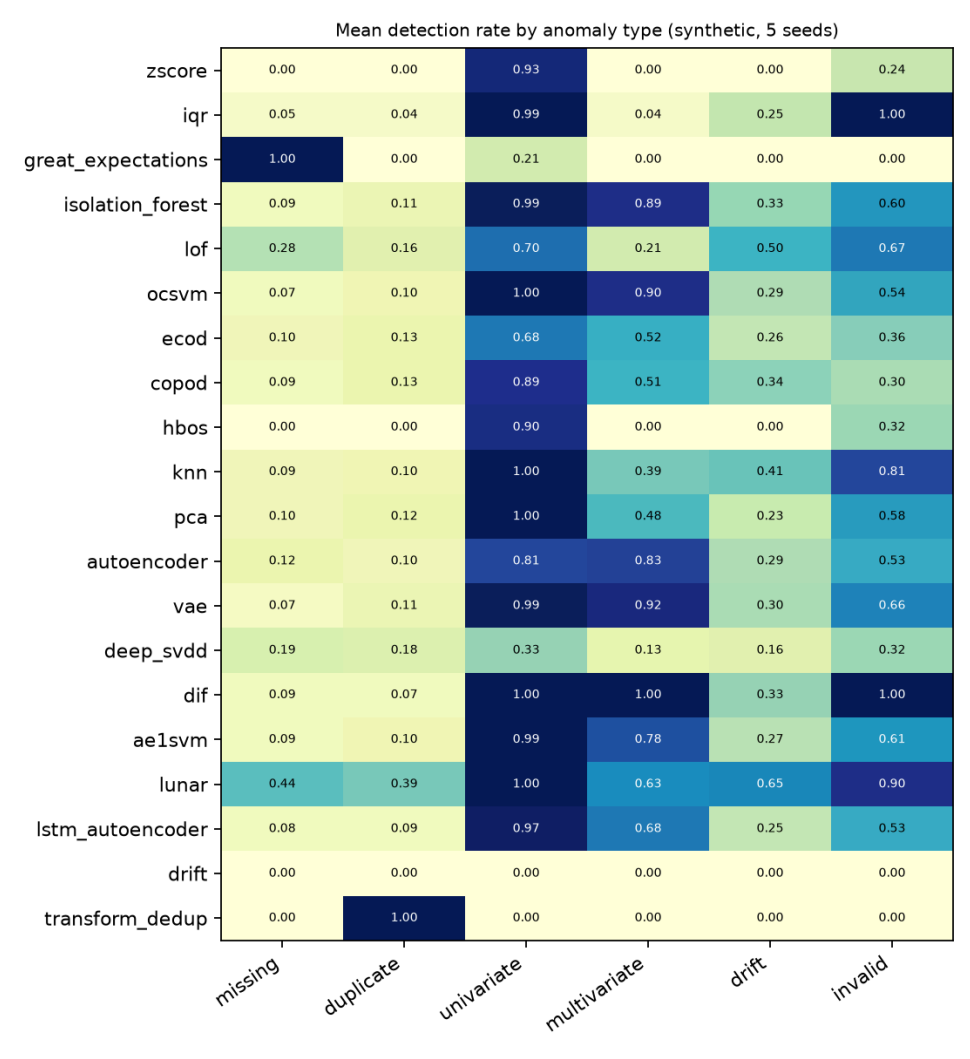
Mean recall over five seeds. Same generator logic for all three schemas: the agreement shows the profile does not depend on one table, not that it holds on real data.

More detectors, same competence map, one exception

Eleven rows added to the synthetic matrix, with three hypotheses written beforehand; none holds. ECOD and COPOD find 0.5 of the multivariate outliers although they read one column at a time, PCA reaches 0.48 against 0.83 for the autoencoder, and LUNAR catches 0.65 of the drifted rows (LOF 0.50) while flagging 41% of clean rows. The LSTM still does not fix drift.

0.65
LUNAR recall on drift, with 41% of clean rows flagged

1.00
DIF recall on multivariate outliers



Detection latency is not pipeline latency

Issue #53. Synthetic matrices from 1,000 to 300,000 rows, a fresh process per measurement so that peak memory belongs to that run, 300 s budget. On the full Credit Card file the Z-score pipeline takes 4.8 s of which detection is 3%, while Isolation Forest takes 8.5 s with detection at 43 to 45%.

133 s
LOF at 300,000 rows (exponent 1.51)

>300 s
One-Class SVM at 100,000 rows

Detector	10k rows	100k rows	300k rows	Memory MB
Z-score	0.002 s	0.013 s	0.039 s	70
IQR	0.004 s	0.029 s	0.086 s	116
Isolation Forest	0.16 s	1.09 s	3.2 s	140
LOF	0.73 s	18.0 s	133 s	405
One-Class SVM	5.6 s	>300 s	not run	788
ECOD / COPOD	0.045 s	0.68 s	2.4 s	186
KNN	0.45 s	11 s	68 s	148
VAE	6.8 s	69 s	209 s	105
DIF	23 s	262 s	>300 s	2,748

Median detection-stage time, 8 features; five of the ten newer detectors shown (paper has all). Autoencoders and the other neural models are in the paper.

31 THE PRACTICAL OUTPUT

Anomaly type to method

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. That itself is a finding a single aggregate ranking would never surface.

Anomaly type	Use	Recall	Why
missing value	Great Expectations	1.00	not-null expectation, exact
exact duplicate	transform dedup stage	1.00	row equality, not a detector's job
univariate outlier	any statistical rule	0.92–0.99	single-column band is enough
multivariate outlier	Isolation Forest	0.91	only the joint models see it (precision 0.24)
temporal drift	windowed monitor	≤0.49	sub-threshold per row; needs a stage
invalid format	IQR, partially	1.00 / none	numeric part caught, string format missed

Unsupervised detection works in one of three regimes

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". Latency and scaling then constrain which detector can sit in a real flow: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

Fraud

too extreme

0.17% positives, and the label mixes every defect. Best F1 of seventeen detectors is 0.35 (Deep SVDD), best precision 0.34 (HBOS).

Cardio

the fit

9.6%, rare and distinct. Isolation Forest F1 0.62, AP 0.70. The regime the detectors are built for.

Water

wrong regime

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance.

The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist further: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

33 WHAT WAS HARDER THAN THE PLAN ASSUMED

Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime, the pipeline cost, matters as much as the algorithm.

THE OPERATING POINT IS A LEVER

Contamination = prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20 and multivariate recall from 0.41 to 0.91.

ONE REGIME OUT OF THREE

Cardio: F1 0.62. Fraud (mixed label): nothing usable. Water Quality (61% majority): at chance.

GX 1.X IS A FULL API REWRITE

Per-row failures need the COMPLETE result format plus an explicit index column. Figured out by probing the installed library.

PYOD SCORES CAN DEPEND ON THE BATCH

ECOD, COPOD and DIF use the scored rows for their statistics, the VAE draws a random code, AE1SVM crashes on a last batch of one row. Recorded adaptations, with 17 row-level batch-independence checks; not upstream-equivalence claims.

ML IS WEAK ON MISSING VALUES

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.08 to 0.26 for the learned models.

DRIFT IS NOT A PER-ROW PROBLEM

Once diluted by clean recent rows the window mean has moved by about a tenth of a standard deviation.

Historical verification has a defined scope

These measurements belong to local run d4304c5. They cover the core pipeline and experiment scripts, with separate verifier checks. They do not certify the new research modules or establish scientific validity. The run is not independently attested.

1,292
tests, plus 3 on full datasets

97.61%
mutants killed (4,980 of 5,102)

Check	Result	Scope
line coverage	1,790 / 1,790	pipeline modules and experiment scripts
branch coverage	350 / 350	same scope
mutation, computing code	95.68%	split, metrics, generator, detectors
mutation, input / output	98.87%	readers, writers, scripts
fault injection	23 / 23	four went through every earlier test
real-data check	pass	three CSV files, manuscript tables match

`Survivors (122) count as failures and were classified by reading the changes, not by proof.`

A surviving mutant is a behaviour no test protects

1 INJECT ONE SMALL BUG

```
- flags = scores > info["threshold"]
+ flags = scores >= info["threshold"]
```

A mutant is a copy of the code with one small, plausible bug. This one is real, in `research/alert_budget.py`: a score equal to the training threshold would now raise an alert.

2 RUN THE TESTS

`killed`	a test fails: the change is noticed
`survived`	every test passes: the behaviour is not checked

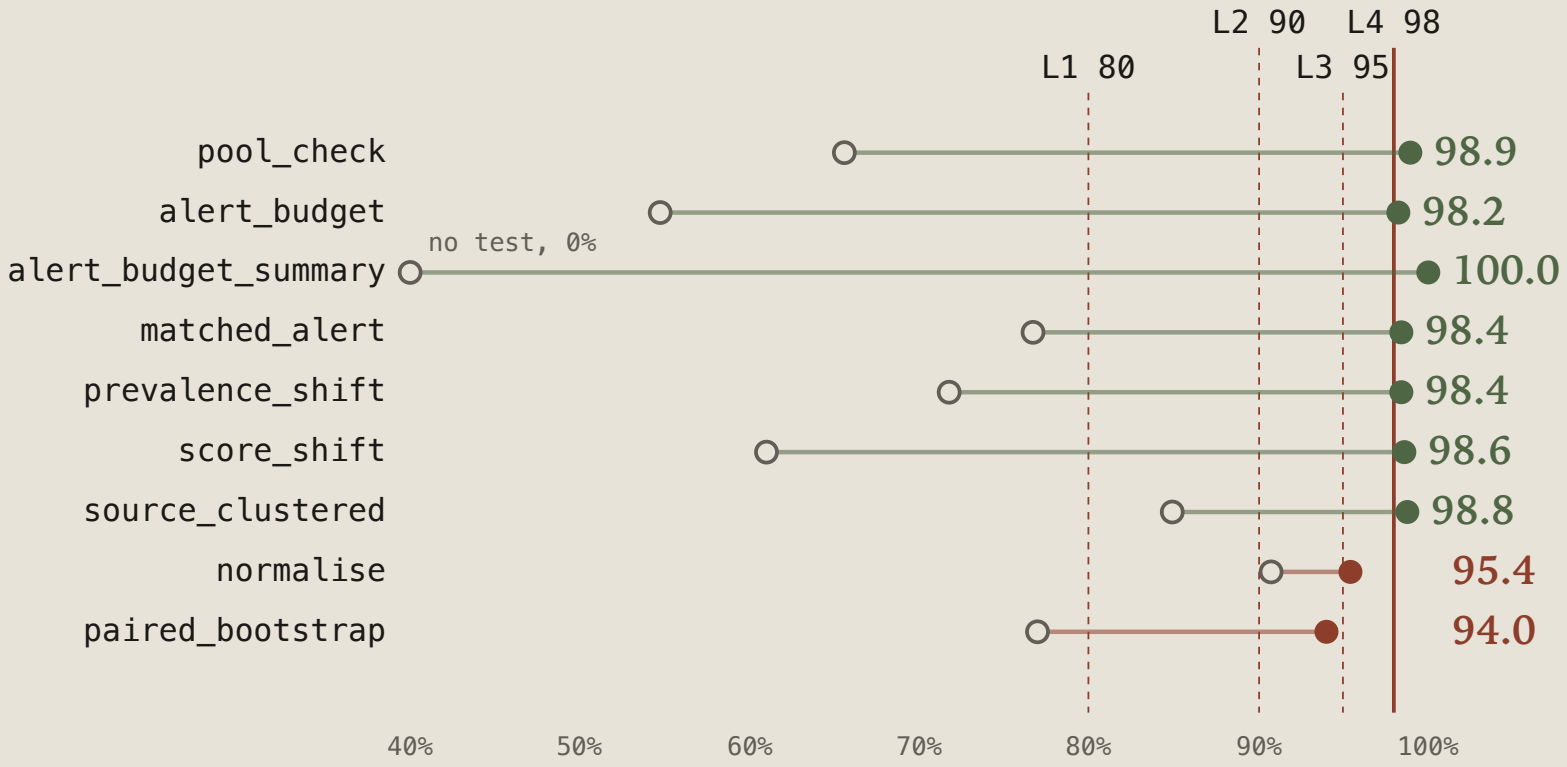
A new test, "a score equal to the training threshold is not an alert", kills this mutant. No test covered it before this session.

3 SCORE THE MODULE

$$\frac{\text{killed}}{\text{eligible mutants}}$$

Example: pool_check killed 178 of 180 eligible mutants, 98.9%. Survivors stay in the denominator. A mutant counts as equivalent only when the owner has recorded it as such, and never by assumption.

Seven of nine modules reach the 98% bar, two do not



Raw focused mutation score per module, local run, 2026-11-11. Open circle: before this session. Filled: after new behavioural tests. Green: at or above the L4 threshold of 98%, on that module alone. All nine are L4, 9 of 82 L4 modules, chosen by the author and tested for these mutants. Axis starts at 40%. Before values: seven from the first run of this session, normalise and paired_bootstrap from the 2026-11-07 campaign. L1 to L3 are shown for reference. Thresholds were not changed.

A focused pass does not certify the repository

LSTM AUTOENCODER, ONE MODULE, ONE SHARD

97.61%
raw, 368 of 377, below the 98% L4 bar

98.66%
adjusted, 368 of 373, author-accepted

Adjusted means four equivalences accepted by the repository owner, as sole author and reviewer, not an independent review. Five inconclusive mutants stay eligible. No timeout.

FIRST COMPLETE NATIVE CAMPAIGN, 2026-11-07, 82 L4 MODULES



65.5% of L4 mutants killed against 98% required, 7,501 further mutants with no covering test. The release gate fails and stays closed.

OPEN
31 L4 modules below threshold and 25 incomplete in that campaign, not re-measured here

OPEN
survivors that look inert (dtype coercions, `zip(strict=)`) await an owner decision

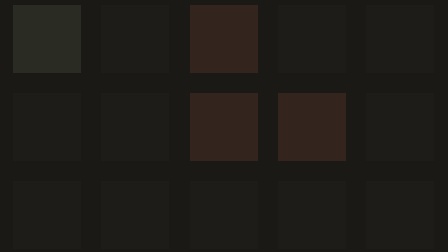
FOUND
minimum anomaly count 3 is below the 6 a stratified split needs; no archived result changes

Experiments run, paper in IMRaD form

Historical implementation issues #63 to #70 are closed. The current roadmap preserves T001–T415. 91 of the 415 tasks are checked as of 2026-11-11, including the exploratory deployment-threshold tasks T163 to T171 of #94; T172, T062 and T063 remain open, as do the issues themselves. Source-aware inference, partition recovery, replay and publication corrections remain required. Optional P2 research is outside the active scope.

held-out split, 5 seeds, chronological and LSTM pilots, cost study	DONE
dataset pool of 152 sets (#66), typed injection on 42 tables (#65)	DONE
benchmark on the 100 labelled pool problems	DONE
deep rows (#64), multi-schema generator (#67)	DONE
historical IMRaD restructuring (#68, #69, #85); final bibliography audit remains #101	DONE
decision-level study under temporal shift (#63)	DONE
paper layout (tables) and discussion revised, merged to main	DONE
complete registry: 17 detectors, every study rerun	DONE
every study extended to the 152-dataset pool, six hypotheses	DONE
P0 remediation: detector/source audits, source-aware statistics, archive replay, publication	NEXT

Rules and machine learning are complementary by anomaly type.



The per-type heatmap is what makes that choice actionable. Across the three example datasets, unsupervised detection pays off only on Cardio, where the anomalies are genuinely rare outliers. On 100 labelled problems of the pool, rankings require train-only revalidation. On 42 real tables with injected types, the synthetic pattern holds only in part, which is why the real-data matrix is the differentiator. The pipeline, the 152-dataset pool, the protocols and the verification layer are in place. The IMRaD write-up and the remaining studies are what remains.